

Linear Models & Least Squares

INFO 521 · Module 1 · Lecture A — Setup, Data, and the Linear Model

Greg Chism

School of Information · University of Arizona

July 2026

Learning outcomes

By the end of Lecture A you should be able to:

1. **Frame supervised regression** in PML notation — inputs, target, and what we are learning.
2. **Load and preprocess** the NHANES data and **state the leakage rule** for choosing features.
3. **Define** the linear model $\hat{y} = \mathbf{w}^\top \mathbf{x}$ and the mean-squared-error loss, and **compute $\mathcal{L}$ by hand**.
4. **Fit and interpret** a scalar least-squares line in clinical units.

The course spine

One clinical outcome — **systolic blood pressure** — seen through **four lenses** this term:

- **Least squares** (Module 1) — fit a line by minimizing squared error.
- **Maximum likelihood** (Module 3) — the same fit, now from a noise model.
- **Bayesian** (later) — carry uncertainty, not just a point estimate.
- **Discovery** (later) — structure and labels we did *not* hand the model.

NHANES recurs in every module so the *methods* change, not the data — you build intuition on one clinical problem.

Supervised regression

Three kinds of machine learning:

- **Supervised** — learn from **labeled** examples $(\mathbf{x}_n, y_n)$. *Regression* = continuous target; *classification* = discrete labels.
- **Unsupervised** — find structure with **no** labels (clustering, PCA) — the term's back half.
- **Reinforcement** — learn from a **reward** signal; out of scope here.

Regression is supervised learning with a continuous target. Given $(\mathbf{x}_n, y_n)$, learn f so $\hat{y} = f(\mathbf{x}) \approx y$ on new $\mathbf{x}$ — here, systolic BP in mmHg.

Notation & data structures

One example is a vector $\mathbf{x}_n \in \mathbb{R}^D \rightarrow$ one prediction $\hat{y}_n = \mathbf{w}^\top \mathbf{x}_n$. Stack the dataset:

$$\mathbf{X} = \begin{bmatrix} \text{---} \mathbf{x}_1^\top \text{---} \\ \vdots \\ \text{---} \mathbf{x}_N^\top \text{---} \end{bmatrix} \in \mathbb{R}^{N \times D}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^N.$$

Rows of the **design matrix** $\mathbf{X}$ are patients; $\mathbf{y}$ is the **target vector**. (We write targets y — not t .)

The data — NHANES 2021–2022

- $N \approx 5,102$ U.S. adults; **target** = systolic blood pressure (mmHg).
- **Six leakage-safe features** (the Project-1 design):

feature	meaning
age	age (years) — the Project-1 primary predictor
bmi	body-mass index
waist	waist circumference
chol	total cholesterol
hdl	HDL cholesterol
hba1c	glycated hemoglobin (HbA1c)

The leakage rule

Exclude any predictor that is itself a blood-pressure measurement — or that would trivially determine the label.

- Diastolic BP (dbp) is **reserved**, never a feature: it co-measures the same physiology as the target.
- This is a **by-design** modeling choice, made *before* fitting — not something the algorithm discovers.

A first look at the data

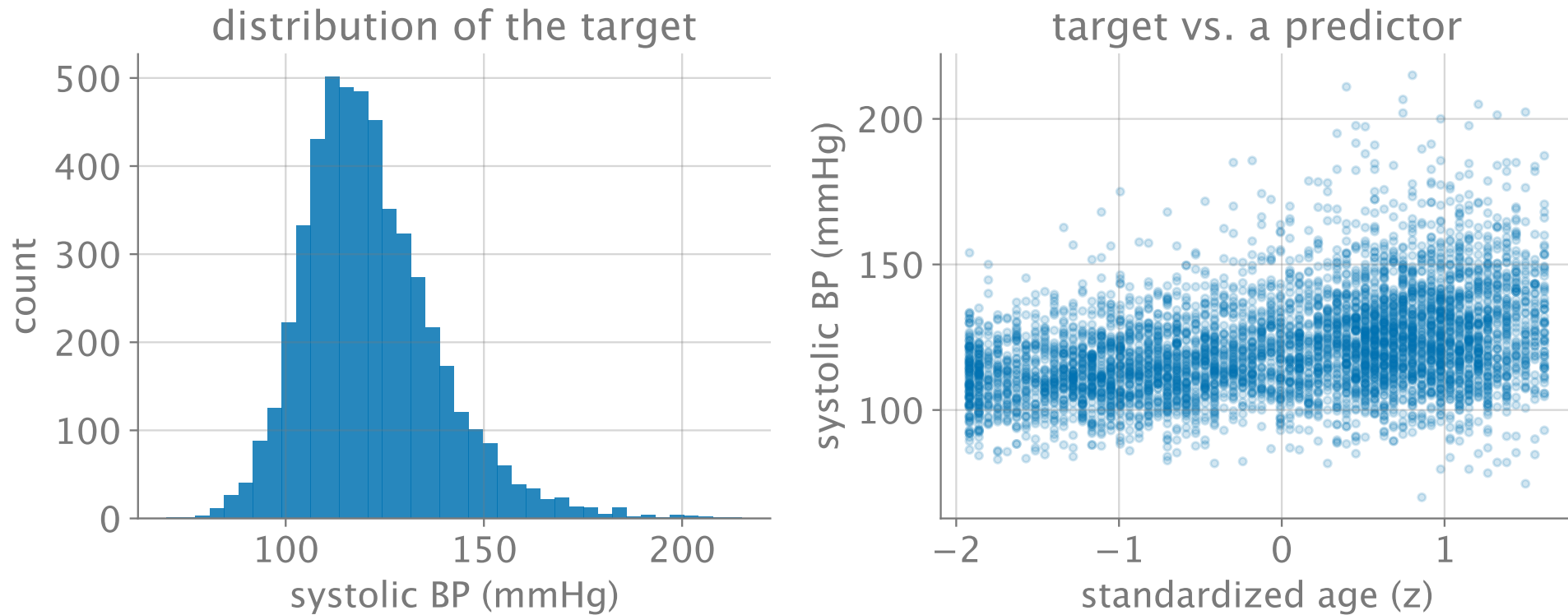


Figure 1: NHANES systolic BP: marginal distribution (left) and vs. standardized age (right).

The linear model

Predict the target as a weighted sum of the features (with an intercept):

$$\hat{y} = \mathbf{w}^\top \mathbf{x} = w_0 + w_1x_1 + \cdots + w_Dx_D.$$

- w_0 — the **intercept** (prediction when all features are 0).
- w_j — the **per-unit effect** of feature j : how $\hat{y}$ changes when x_j rises by one unit, holding the rest fixed.

The loss: mean squared error

Measure fit by the average squared miss:

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N (y_n - \mathbf{w}^\top \mathbf{x}_n)^2 = \frac{1}{N} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2.$$

Why **squared** error? It is **smooth** (differentiable — we'll set its gradient to zero), and it **penalizes large misses** disproportionately.

A squared loss is also exactly what a Gaussian-noise model gives you — that tie is Module 3.

Worked: compute $\mathcal{L}$ by hand

Toy data (x_n, y_n) : $(0, 1)$, $(1, 1)$, $(2, 3)$, with a guess $\hat{y} = 0.5 + x$.

Predictions $\hat{y}_n$: 0.5, 1.5, 2.5.

Residuals $y_n - \hat{y}_n$: 0.5, -0.5, 0.5.

Square & average: $\mathcal{L} = \frac{1}{3}(0.5^2 + (-0.5)^2 + 0.5^2) = \frac{1}{3}(0.75) = 0.25$.

A scalar least-squares fit

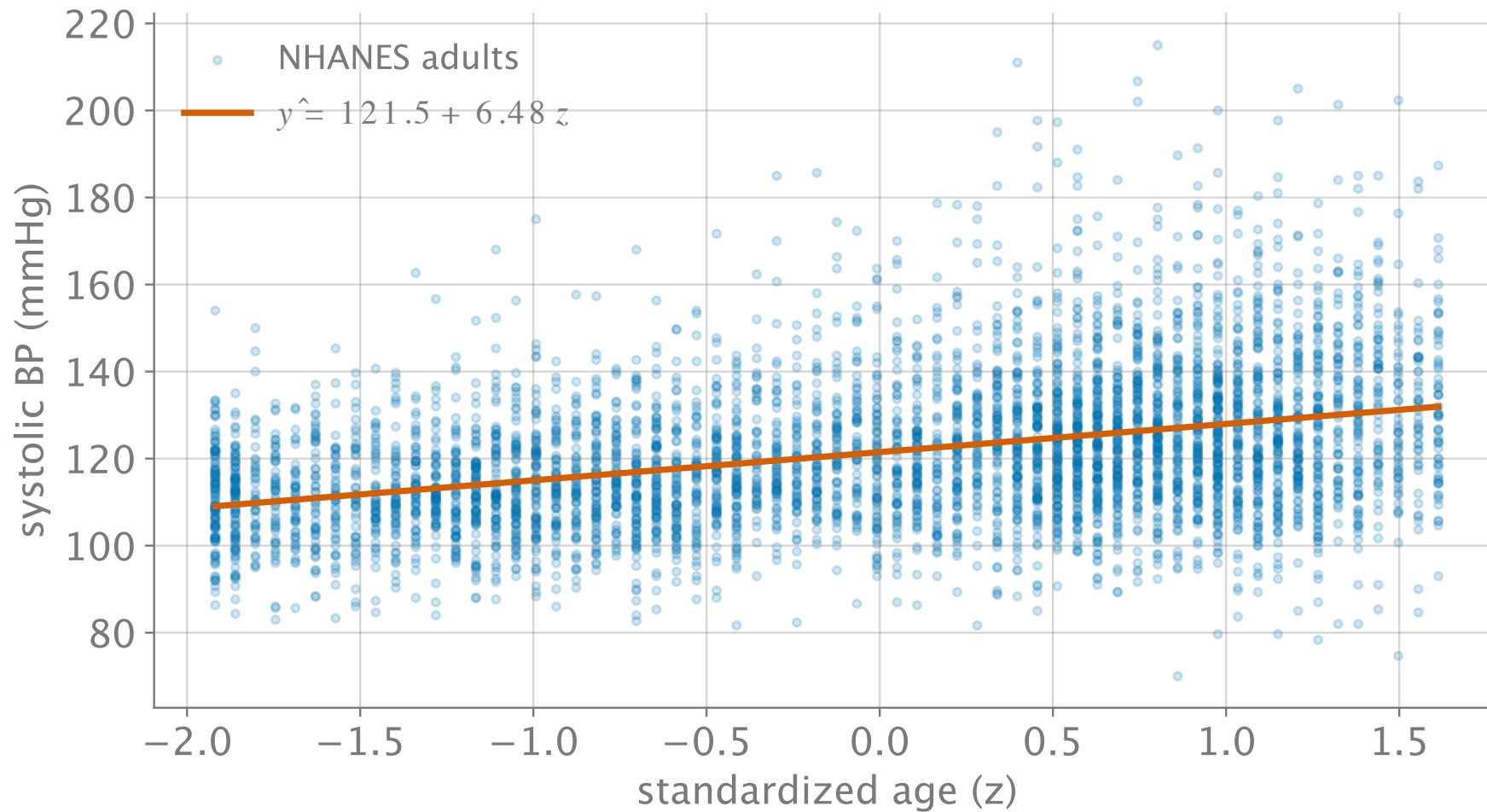


Figure 2: Least-squares fit of systolic BP on standardized age.

Interpreting the fit

The fitted line is $\hat{y} = w_0 + w_1 z$ with $w_0 \approx 121.5$, $w_1 \approx 6.5$ (mmHg per **standard deviation** of age).

- **Intercept** w_0 : predicted SBP at *average* age (where $z = 0$) — about **121.5 mmHg**, a clinically sensible central value.
- **Slope** w_1 : +1 SD of age (≈ 17 years) adds ≈ 6.5 mmHg, i.e. ≈ 0.38 **mmHg per year**.
- **Sanity check** — sign is positive (BP rises with age) and the magnitude is modest and plausible. Always translate weights back to units.

Live demo: the least-squares explorer

The linear model: least squares by hand

Each point is a patient: age on the x-axis, resting systolic blood pressure on the y-axis. Drag the endpoints of the line, or use the sliders, to fit it by eye. Watch the residuals and the loss respond, then compare your fit to the least-squares optimum.

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N \text{bigl}(y_n - (w_0 + w_1 x_n)\bigr)^2$$

minimized by

$$\hat{w} = (X^{\text{top}} X)^{-1} X^{\text{top}} y$$

Patients Your fit Least squares Residuals Outlier

Intercept w_0

Slope w_1

Show least-squares solution

Add outlier

Reset

Intercept w_0

—

Slope w_1

—

Your loss (MSE)

—

Least-squares MSE

Explorer — static fallback

The iframe above does not print or export to PDF; this snapshot stands in.

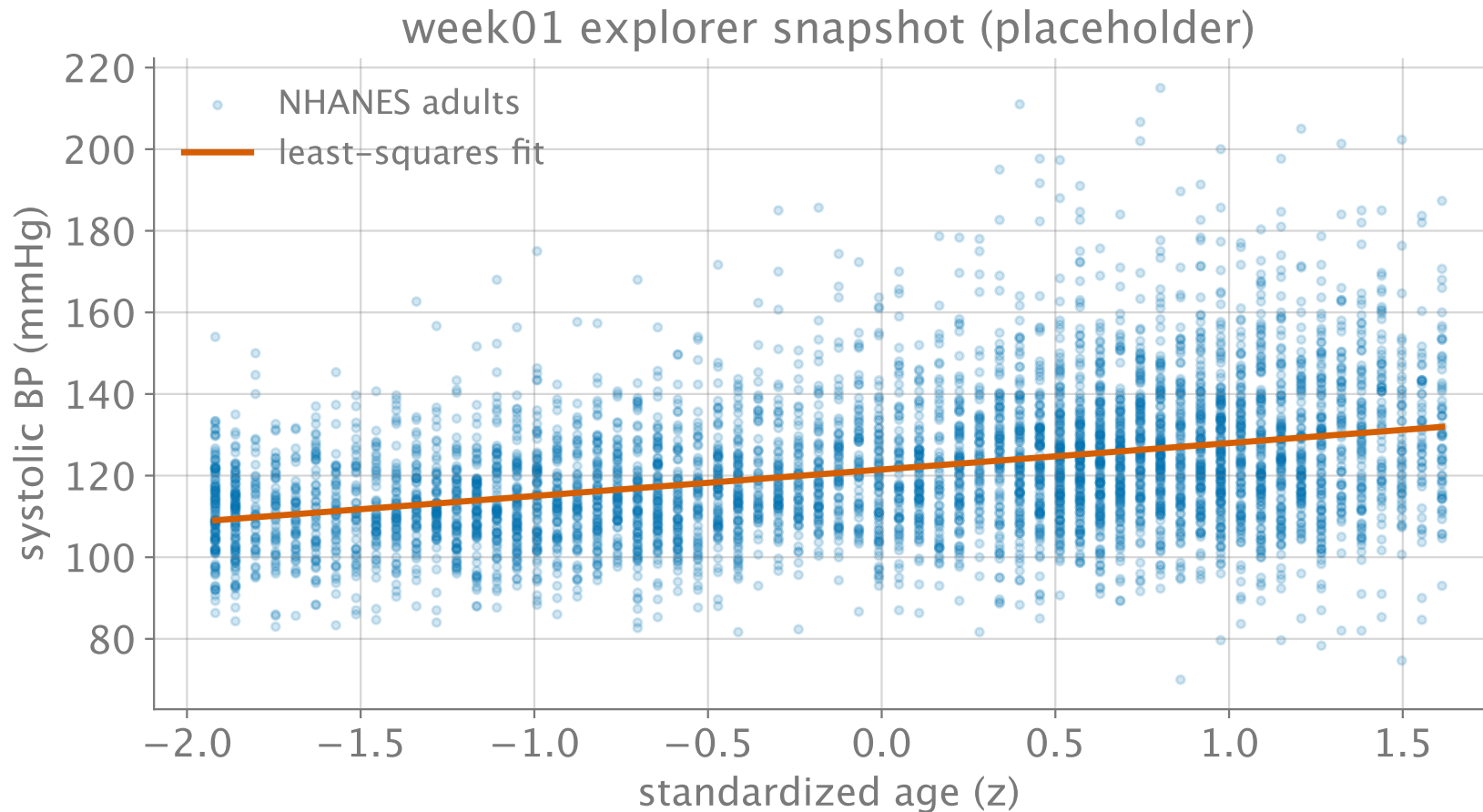


Figure 3: Static fallback snapshot of the week01 least-squares explorer.

Recap & bridge

We now have the two ingredients of least squares:

- a **model** — $\hat{y} = \mathbf{w}^\top \mathbf{x}$, and
- a **loss** — $\mathcal{L}(\mathbf{w}) = \frac{1}{N} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$.

Next (Lecture B): solve for the best $\mathbf{w}$ in closed form — set the gradient to zero — and see the geometry behind it.